
How to Pick the Model: A Framework for Budget-Aware Orchestration

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Abstract

1 Agent systems on verified tasks invite a deceptively simple question: which back-
2 bone model should be run? The strongest model is not always the best deployment
3 choice, because verified progress must be bought under finite wall-clock, token,
4 API, and evaluation budgets.

5 We formulate this problem as deployment-aware model selection for verified
6 agent systems. A fixed router observes a pre-deployment information state and
7 chooses a worker or harness action; the decision is scored by a checker-indexed
8 deployment loss rather than by routing accuracy or final checker score alone.
9 The key estimand is paired: on the same task instances, with the same router
10 and action menu, does richer information change the selected action and reduce
11 net deployment loss after its own measurement and context costs are charged?
12 We instantiate the framework on a Karpathy AutoResearch-inspired CIFAR-10
13 edit-verify benchmark, where a prompt-only router chooses among heterogeneous
14 workers before costly deployment runs.

15 The protocol separates three objects that benchmark racing conflates: worker-
16 frontier structure, information availability, and deployable routing value. The main
17 two-worker study shows that first-hit speed, final improvement, log-effort, and
18 deployment loss can rank workers differently. The same diagnostics then locate a
19 routing failure: probe signals expose the task modes, but the evaluated prompt-only
20 router does not convert that information into positive paired deployment gain.

21 The resulting protocol evaluates not which agent is best in isolation, but whether
22 measured information is actually usable by a router under the charged deployment
23 objective.

24 1 Introduction

25 Practitioners building agent systems face multiple practical challenges: which model (backbone) to
26 use, how to route between models, when to spawn sub-agents, how to engineer context between them,
27 and so on. All these possible choices often affect performance (quality of the solution), speed (time
28 to solution), and token/API cost. The question is how to choose among the possible designs. The
29 common way to answer this question is benchmark racing: take a benchmark such as SWE-bench
30 [1] or MLE-bench [2] and pick the agent system with the best average end-to-end performance. An
31 increasingly long line of work based on this recipe is proposing better and better agentic system
32 architectures to achieve state-of-the-art performance on these tasks. This article investigates

33 *whether there are other ways to answer the question of which agent system is most suitable.*

34 Indeed, there are many tasks for which such benchmarks do not exist, such as developing academic
35 research [3]. Moreover, tasks often contain instances of varied difficulty and the practitioner may

want to understand whether, on a single instance, a lower-cost, faster model is sufficient; the system should be given more tools; or a more involved orchestration is required. As a running example for the rest of the paper, the question of whether a user should use Haiku, Sonnet, or Opus in Claude Code, or Claude Code rather than Codex, for their own task is not fully addressed by benchmark racing.

The central issue is that one benchmark score may conflate mechanisms that imply different engineering decisions. A system can win because its worker has lower per-step time or token cost, or because the selected worker is more competent on a solver-relevant task mode. Analogously, imagine a pre-deployment signal (the problem definition, context, or an initial test) reveals information about the instance that can be used to choose a better backbone or orchestration. A system can win by exploiting that information when choosing a model or orchestration, or it can lose by failing to exploit it—as humans implicitly do when they decide to use lower or higher reasoning effort or different models on their tasks.

More practically, in this article *we study the deployment problem of choosing which model or harness to run on a verifiable task*. The objective is time to reach a required verified result: among actions that may eventually succeed, the useful one is the one expected to reach the verifier threshold fastest, after charging any pre-deployment measurement used to make the choice.

Indeed, Achille and Soatto [4] recently argued that time to solution, as formalized by their Proper Time, is a performance measure for comparing agents on verifiable tasks.

Building on this intuition, we operationalize a certified log-effort surrogate gap between two possible designs. We show that it mathematically decomposes into four components that isolate the mechanisms of choice mentioned above.

The experimental protocol estimates the operational quantities that are directly measurable and tests, by paired selected-worker deployment, whether richer pre-deployment information improves net deployment loss after measurement costs are charged. This gives the paper’s main diagnostic separation: first estimate whether a worker frontier contains mode-conditional structure, then test whether pre-deployment records reveal that structure, and finally validate whether the deployed router turns the information into positive paired deployment gain.

Research questions and contributions. We organize the paper around three questions, each paired with the contribution that answers it.

1. **Can the certified log-effort surrogate gap between agent designs be decomposed into meaningful terms?** Theorem 1 gives an exact four-term identity under a packed latent-mode assumption: cost, within-mode competence, routing information, and a predeclared mode-summary divergence additively explain the improvement in expected certified log-effort.
2. **Can the decomposition separate frontier structure, information availability, and deployable routing value?** Algorithm 1 and the pilot concentration bound in Appendix A turn the identity into a structured pilot protocol that estimates measurable worker-side terms, audits router-visible signals, and reserves deployment recommendations for paired selected-worker validation.
3. **When does a lower-cost model with retries beat a stronger model?** Theorem 2 gives the single-mode crossover rule, and the multi-mode accounting shows why the answer changes across task modes.

The experiments below instantiate these claims with checked trajectories and deployment accounting.

2 Setup and accounting

We study the problem faced by a deployment system before launching an agentic run: given a task instance and a finite menu of possible workers or harnesses, which one should be run? The answer is not necessarily the model with the best final score. A deployment action must produce checked progress under finite time, token, API, and evaluation budgets.

Definition 1 (Budgeted externally checked task). *A budgeted externally checked task is a tuple*

$$\mathcal{T} = (\mathcal{X}, \mathcal{Y}, V, B_{\text{dep}}, \mu).$$

84 Here $\mathcal{X}$ is an instance space, $\mathcal{Y}$ is a solution space, $V : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{R}$ is an external checker or
 85 scoring procedure on instance–artifact pairs, B_{dep} is the full deployment budget and acceptance
 86 object, and μ is the deployment distribution over instances. The object B_{dep} may include a horizon,
 87 verifier budget, token/time/cost caps, and a thresholding rule that induces a binary success event
 88 from the checker score. For a realized instance $x \in \mathcal{X}$, an agent produces artifacts $y \in \mathcal{Y}$ that are
 89 scored by $V(x, y)$. An agent succeeds on $x \sim \mu$ if it produces an accepted y before exhausting B_{dep} .

90 The checker may be binary, as in unit tests or proof checking, or scalar, as in validation loss or reward.
 91 The key point is externality: progress is determined by the checker, not by the model’s self-report.

92 **Definition 2** (Workers, actions, and routers). Let $\mathcal{M} = \{M_1, \dots, M_K\}$ be a finite menu of workers,
 93 where a worker is any deployable solver or fixed agent policy that can produce a candidate solution
 94 for verification. Let $\mathcal{A}$ be a finite menu of deployable actions. An action may be a single model, a
 95 fixed model harness, a retry policy, or a tool-using agent protocol. In the simplest setting, $\mathcal{A} = \mathcal{M}$;
 96 more generally, an action bundles a worker with fixed stopping, retry, tool-use, and carryover rules.

97 The deployment loss is defined in Section 4, where it is tied to first-passage cost, audit loss, and
 98 measurement loss. A routed design is a tuple

$$d = (\mathcal{A}, R, \kappa),$$

99 where R is a router that selects an action after observing permitted information and κ is the declared
 100 scalar action-cost convention used in the log-effort diagnostic. When costs are logged as resource
 101 vectors, the protocol must fix the scalar reduction before evaluation. A router R observes a permitted
 102 pre-deployment information state $Z_j(x)$ and selects an action

$$a_j(x) = R(Z_j(x)) \in \mathcal{A}.$$

103 Equivalently, a signal condition induces the routing policy $\rho_j : x \mapsto a_j(x)$. The selected action then
 104 performs the deployment run from the original task state.

105 This separates routing from deployment. The router chooses what to run; the selected action performs
 106 the actual checked run. Unless explicitly declared otherwise, pre-deployment records are shown only
 107 to the router and are not passed to the selected worker.

108 **Definition 3** (Task instances and task modes). Given $\mathcal{T} = (\mathcal{X}, \mathcal{Y}, V, B_{\text{dep}}, \mu)$, a task instance is
 109 a concrete realization $X = x \in \mathcal{X}$ of that same task. It is the measurable object supplied to the
 110 deployed system; the router and worker may access only the fields made available by the protocol. A
 111 task mode is a latent variable

$$S = \sigma(X) \in \mathcal{S} = \{1, \dots, m_S\}$$

112 induced by a measurable map $\sigma : \mathcal{X} \rightarrow \mathcal{S}$ that assigns each instance to one of finitely many solver-
 113 relevant variants of the task. It defines a prior $\pi_s = \mathbb{P}_\mu[S = s]$ and mode-conditional instance
 114 distributions $\mu_s = \mu(\cdot \mid S = s)$. Modes are therefore not different tasks and not worker labels. They
 115 are variants of the same task: finite subpopulations of instances that share solver-relevant structure
 116 and may favor different workers or interventions.

117 The evaluator retains S only for accounting: neither the router nor the worker needs to observe it. In
 118 the experiments, each mode is represented by one fixed executable setting, so the reported quantities
 119 are fixed-panel analogues rather than population-level estimates under μ .

120 **Assumption 1** (Packed latent-mode family). Each instance $X \sim \mu$ is associated with a mode
 121 $S = \sigma(X)$ in a finite mode set $\mathcal{S}$, with prior π . Conditional on mode s , action a has success
 122 probability $p(a, s) > 0$ and scalar cost

$$\kappa(a, s) = \mathbb{E}[C_\delta(a, X; \xi) \mid S = s] > 0$$

123 under the declared deployment protocol, where C_δ is the normalized cost accumulated to the first
 124 certified success or the horizon. At deployment time, the router observes a signal $Z = z$ and selects
 125 one action. The evaluator defines the mode distribution induced by that signal,

$$\pi_z(s) = \mathbb{P}[S = s \mid Z = z].$$

126 The router is not required to output π_z or any mode distribution. Instead, the theory uses a fixed
 127 evaluator-side mode-summary distribution $q_z \in \Delta(\mathcal{S})$, with $q_z(s) > 0$ whenever $\pi_z(s) > 0$. This

summary is not a router output; it is a theory-facing diagnostic distribution specified before evaluation. Write $a_z = \rho(z)$ for the action selected after observing z . For an instance whose true mode is s , the certified effort surrogate is proportional to

$$T_{\rho,q}(s, z) \propto \frac{\kappa(a_z, s)}{p(a_z, s) q_z(s)}. \quad (1)$$

Assumption 1 is a diagnostic abstraction, not an empirical claim that the finite schema is universally correct. It isolates the regime where cost, competence, signal information, and mode-summary mismatch are algebraically separable; residual, calibration, and held-out deployment diagnostics test whether this abstraction is useful for the declared panel.

Definition 4 (Cost-adjusted certified log-effort diagnostic objective). For a fixed routed policy $\rho : Z \mapsto a_Z$ and a fixed evaluator summary q satisfying Assumption 1, define

$$\mathcal{J}(\rho, q) = \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} [\log \kappa(a_Z, S) - \log p(a_Z, S)] + \mathbb{E}_Z [\mathcal{H}(\pi_Z) + \text{KL}(\pi_Z \| q_Z)]. \quad (2)$$

Lower $\mathcal{J}$ means lower expected certified log-effort under the fixed diagnostic convention. The quantity q is not optimized by the router; it is fixed by the evaluator-side diagnostic rule. When several signal conditions are compared, $\mathcal{J}$ is applied to the corresponding policy summaries; Appendix C writes this comparison explicitly for Z_j versus Z_0 .

The action quantities in (2) are $\kappa(a_Z, s)$ and $p(a_Z, s)$: how costly the selected action is on mode s and how often it succeeds there. The router and information quantities are Z , π_z , and q_z : what the router is allowed to observe, what that signal reveals about the task mode, and the evaluator-side mode-summary distribution used in the KL term.

Remark 1 (Meaning of the effort surrogate). Equation (1) says that, on a true mode s after signal z , certified effort increases with action cost, decreases with action success probability on that mode, and increases when the evaluator-side mode summary gives too little diagnostic mass to that mode. The router is not literally routing jobs to modes; modes are evaluator-defined task variants. The quantity $q_z(s)$ is a diagnostic summary, not a deployment probability. The KL term below is therefore a mode-summary divergence under the chosen diagnostic convention. It should be interpreted as a router failure only if the experiment predefines a reproducible construction of q_z from router behavior.

3 Accounting identity

For a fixed signal value z , the routing part of the objective is the cross-entropy between the signal-induced mode distribution and the fixed evaluator-side mode-summary distribution:

$$\text{CE}(\pi_z, q_z) = - \sum_{s \in \mathcal{S}} \pi_z(s) \log q_z(s) = \mathcal{H}(\pi_z) + \text{KL}(\pi_z \| q_z). \quad (3)$$

The KL term is a divergence over finite task-mode distributions. It is not a token-level language-model divergence.

Theorem 1 (Four-term routed-policy decomposition). Under Assumption 1, compare a baseline policy ρ_0 that deploys fixed action a_0 without using Z and uses $q_0(\cdot | z) \equiv \pi$, with a deployed routed policy summarized by (ρ, q) . The improvement in expected certified log-effort, $\Delta = \mathcal{J}(\rho_0, q_0) - \mathcal{J}(\rho, q)$, decomposes as

$$\Delta = \underbrace{\mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[\log \frac{\kappa(a_0, S)}{\kappa(a_Z, S)} \right]}_{\text{cost}} + \underbrace{\mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[\log \frac{p(a_Z, S)}{p(a_0, S)} \right]}_{\text{competence}} + \underbrace{\mathbb{I}_\pi(S; Z)}_{\text{routing information}} - \underbrace{\mathbb{E}_Z [\text{KL}(\pi_Z \| q_Z)]}_{\text{mode-summary divergence}}. \quad (4)$$

Proof sketch. Taking logs in (1) gives $\log T_{\rho,q}(s, z) = C + \log \kappa(a_z, s) - \log p(a_z, s) - \log q_z(s)$. Conditioning on $Z = z$ and averaging over $S \sim \pi_z$ converts $-\mathbb{E} \log q_z(S)$ into the cross-entropy in (3). Subtracting the prior-matched baseline cancels constants and uses $\mathcal{H}(\pi) - \mathbb{E}_Z \mathcal{H}(\pi_Z) = \mathbb{I}_\pi(S; Z)$. The full algebra is given in Appendix A. $\square$

The four terms have different operational meanings. If the cost term is large and positive, a lower-cost action may be preferable even when it reaches success later in step count. If the competence term varies by mode, a global worker ranking is likely to hide comparative advantage. If information gain is high, the pre-deployment signal exposes solver-relevant structure. If the mode-summary divergence is high, the signal-induced mode distribution is poorly matched to the evaluator’s predeclared mode summary. For two signal conditions, the identity compares the routed policies induced by those signals. A richer signal is useful only if its induced action change improves the cost–success tradeoff after the cost of obtaining the signal is charged.

Operational reading. The task mode S is not an oracle label for the best worker. The evaluator fixes a mode schema before evaluation, then estimates the mode-conditional quantities $p(a, s)$ and $\kappa(a, s)$. Routing information asks whether a signal makes those modes more identifiable. Routing shift asks whether the selected action changes when the signal is enriched. Mode-summary divergence is the KL term defined by the fixed evaluator-side mode summary q . Allocation regret, used later in the empirical protocol, is a separate action-frontier diagnostic rather than an estimator of this KL term. The final decision is not made by the information term alone; it is made by paired deployment gain on held-out instances after measurement and context costs are charged. Appendix B gives the full operationalization of these diagnostics.

3.1 Retry crossover design rule

Theorem 2 (Single-mode retry crossover). *Fix a single homogeneous mode, success threshold δ , and horizon H . Let actions h and ℓ have one-attempt first-passage success probabilities $p_h = \mathbb{P}[\tau_\delta(h, X) \leq H]$ and $p_\ell = \mathbb{P}[\tau_\delta(\ell, X) \leq H]$, with expected first-passage costs κ_h and κ_ℓ . Assume $0 < p_\ell < p_h < 1$ and $\kappa_\ell < \kappa_h$. Assume retries of ℓ are conditionally i.i.d. fresh attempts, each with the same one-attempt horizon H and common Bernoulli success probability p_ℓ . The lower-cost retry depth that matches the strong action’s one-attempt success probability is*

$$D_{\text{cross}} = \frac{\log(1 - p_h)}{\log(1 - p_\ell)}.$$

The corresponding integer fixed-depth retry block is cost-favorable whenever

$$\lceil D_{\text{cross}} \rceil \kappa_\ell < \kappa_h.$$

Remark 2 (Stopping after first retry success). *If retries of the lower-cost action stop after the first success and each attempted retry has unconditional expected cost κ_ℓ , the expected cost at depth D is*

$$\kappa_\ell \sum_{d=1}^D (1 - p_\ell)^{d-1} = \kappa_\ell \frac{1 - (1 - p_\ell)^D}{p_\ell},$$

and the decision should also account for the residual failure probability under the deployment loss. Any difference in within-attempt time-to-success is already absorbed into κ_h and κ_ℓ .

The multi-mode case is where the four-term identity matters. A lower-cost worker can be rational on one task mode and irrational on another. A router is useful only when different actions win the cost–success tradeoff on different modes and the pre-deployment information helps identify which mode applies.

4 Deployment validation protocol

The log-effort decomposition guides diagnostics, but deployment recommendations are made only after paired validation under a predeclared loss. The empirical protocol is compact: freeze modes, losses, cost normalizers, and signal records; estimate mode-conditional worker success and first-passage cost; audit whether richer records expose modes and change allocation; then validate finalists by paired selected-worker deployment after measurement cost.

The log-effort objective $\mathcal{J}$ is a diagnostic surrogate, not the deployment loss.

Diagnostic versus decision loss. $\mathcal{J}$ explains cost–competence and signal/accounting structure under a declared diagnostic convention. ℓ_{dep} is the decision-facing loss used for paired deployment validation. We use $\mathcal{J}$ to diagnose why rankings differ, and ℓ_{dep} to decide whether a routed policy actually improves deployment value.

206 For deployment validation we use a predeclared first-passage loss. Given a checked progress trace
 207 $G_1, \dots, G_H$, define

$$\tau_\delta(a, x; \xi) = \inf\{t \leq H : G_t(a, x; \xi) \geq \delta\}, \quad F_\delta(a, x; \xi) = \mathbf{1}\{\tau_\delta(a, x; \xi) = \infty\},$$

208 and

$$\text{Occ}_{\delta, H}(a, x; \xi) = H^{-1} \sum_{t=1}^H \mathbf{1}\{G_t(a, x; \xi) \geq \delta\}.$$

209 Let $c_\delta(a, x; \xi)$ be the normalized worker-side resource vector accumulated to $\tau_\delta \wedge H$, and let

$$C_\delta(a, x; \xi) = \lambda^\top c_\delta(a, x; \xi)$$

210 be the scalar cost used in $\kappa(a, s)$. The primary loss is

$$\ell_{\text{dep}}(a, x; \xi) = \alpha_{\text{fail}} F_\delta(a, x; \xi) + C_\delta(a, x; \xi), \quad (5)$$

211 with $\alpha_{\text{fail}} = 1$ and normalized resource weights λ fixed before evaluation. This is the loss used
 212 in $\mathcal{J}_R(j)$ and $\Delta_R(j)$ unless a different primary loss is explicitly declared. Because the benchmark
 213 records full trajectories to horizon H , we also report the full-horizon audit loss

$$\ell_{\text{audit}}(a, x; \xi) = \ell_{\text{dep}}(a, x; \xi) + \alpha_{\text{occ}}(1 - \text{Occ}_{\delta, H}(a, x; \xi)) + \alpha_{\text{qual}}\psi(G_H(a, x; \xi)), \quad (6)$$

214 where $\alpha_{\text{occ}} = \alpha_{\text{qual}} = 1$ and $\psi(u) = 1 - u$ after clipping $G_H \in [0, 1]$. The audit loss checks
 215 persistence and final quality; it does not change the first-passage cost convention in Theorem 1.

216 **Algorithm 1** (Theory-to-deployment selection protocol).

Input: Task family, checker, finite task-mode schema $\mathcal{S}$, action menu $\mathcal{A}$, admissible progressive signals $\{Z_j\}$, pilot cell count n_{cell} , deployment repeats Q_{dep} , confirmation budget.

Output: Recommended signal-conditioned deployment policy, reported accounting terms, paired deployment gain, and calibration diagnostics.

- 1 *Freeze the modes and loss.* Declare $\mathcal{S}$, success threshold δ , horizon H , cost normalizers, deployment loss (5), measurement costs, and router output contract.
- 2 *Structured pilot.* For each action $a \in \mathcal{A}$ and mode $s \in \mathcal{S}$, estimate $\hat{p}(a, s)$, $\hat{\kappa}(a, s)$, first-hit curves, occupancy, and uncertainty intervals.
- 217 3 *Signal and router diagnostics.* For each progressive signal Z_j , report any predeclared signal-information diagnostic and log the action allocation induced by the router.
- 4 *Accounting score.* Compute the four log-effort terms from Theorem 1 only for quantities whose diagnostic estimators were predeclared; otherwise report allocation regret separately.
- 5 *Paired deployment validation.* On the same holdout instances, run the selected action under Z_0 and richer Z_j conditions, score repeated selected-worker trajectories by (5), and subtract measurement cost.
- 6 *Decision.* Choose the signal condition with the largest held-out $\hat{\Delta}_R(j)$ subject to predeclared gates; use $\mathcal{J}$, frontier scores, and allocation regret as diagnostics, not as the final deployment objective.

218 4.1 Progressive signals and paired router value

219 Let $Z_j(X; \zeta_j)$ be the router-visible pre-deployment record under signal condition j , where ζ_j denotes
 220 measurement randomness. Let Z_0 be the static signal observed before choosing a deployment action.
 221 For any admissible richer condition j , the router observes Z_j , pays the corresponding measurement
 222 and router-context cost, and selects an action

$$a_R^j(X) = R(Z_j(X; \zeta_j); U_R) \in \mathcal{A}.$$

223 Here U_R denotes router-side randomness; for deterministic routers it is fixed and suppressed. Thus
 224 the experimental comparison is between routing policies ρ_0, ρ_j induced by different signals for the
 225 same orchestrator R , not between different orchestrators. The normalized measurement loss is

$$\ell_{\text{meas}}(j, x; \zeta_j) = \lambda_{\text{meas}}^\top d_j(x; \zeta_j),$$

226 where d_j contains measurement wall-clock, tokens, checker calls, context expansion, parsing, retry,
 227 and latency costs. Set $\ell_{\text{meas}}(0, x) = 0$. The weights λ_{meas} use the same predeclared normalizers

as deployment resource cost unless a separate router-cost scale is declared. In the main protocol, pre-deployment probes are outside the selected action’s deployment run: they are charged through ℓ_{meas} and do not reduce the later horizon H or verifier budget B_V .

The selected-action deployment objective is

$$\mathcal{J}_R(j) := \mathbb{E} \left[\ell_{\text{meas}}(j, X; \zeta_j) + \ell_{\text{dep}}(a_R^j(X), X; \xi) \right]. \quad (7)$$

The expectation averages over task instances, measurement randomness, router randomness, and deployment randomness unless a conditional estimand is explicitly declared. The router-facing value of signal condition j is the paired gain

$$\Delta_R(j) := \mathcal{J}_R(0) - \mathcal{J}_R(j) = \mathbb{E} \left[\ell_{\text{dep}}(a_R^0(X), X) - \ell_{\text{dep}}(a_R^j(X), X) - \ell_{\text{meas}}(j, X) \right]. \quad (8)$$

Positive $\Delta_R(j)$ means that the richer signal improves net deployment loss for the same router on the same task distribution. Allocation shift alone is insufficient:

$$\text{Shift}_R(j) = \Pr[a_R^j(X) \neq a_R^0(X)]$$

measures decision relevance, not benefit.

Proposition 1 (Unbiased paired router-gain estimator). *Assume task instances are sampled i.i.d. from μ , the orchestrator, action menu, and signal protocols are fixed before evaluation, measurement costs are recorded without bias, and deployment seeds are independent conditional on selected action and instance. The router is run once per instance and signal condition; the Q_{dep} deployment repeats estimate the conditional expected loss of the selected action. For condition j , estimate*

$$\hat{\Delta}_R(j) = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{Q_{\text{dep}}} \sum_{q=1}^{Q_{\text{dep}}} \ell_{\text{dep}}(a_{R,i}^0, x_i; \xi_{i,q}^0) - \frac{1}{Q_{\text{dep}}} \sum_{q=1}^{Q_{\text{dep}}} \ell_{\text{dep}}(a_{R,i}^j, x_i; \xi_{i,q}^j) - \hat{\ell}_{\text{meas}}(j, x_i) \right].$$

If the repeated-run averages are unbiased for the expected deployment losses of the selected actions, then $\mathbb{E}[\hat{\Delta}_R(j)] = \Delta_R(j)$.

When the same worker is selected under Z_0 and Z_j , using common deployment seeds preserves unbiasedness and reduces paired variance. When counterfactual outcomes for all actions are logged on the same instance, hindsight action regret can be reported as a diagnostic, but it is not the primary estimand. If the study fixes a finite panel of executable settings rather than sampling i.i.d. instances from μ , the same estimator describes that panel and should not be interpreted as a population-level deployment estimand.

5 Experiments and analysis

Setup. We instantiate the framework on a repeated-trajectory panel of fixed executable settings for CIFAR-10 edit-verify optimization. The task modes are three starting optimization variants of the same checked task, $\mathcal{S} = \{\text{MLP-flat}, \text{compact CNN}, \text{micro ResNet}\}$, with empirical weights $\hat{\pi}(s) = 1/3$. They are task modes because the dataset, checker, horizon, and success rule are fixed while the editable artifact’s starting architecture and parameterization change, letting us test whether worker choice is mode-conditional rather than a single global ranking. The analyzed worker menu is $\{\text{gpt_5_3_codex}, \text{gpt_5_4}\}$. Each action is one worker under the common prompt and harness, so $\mathcal{A} = \mathcal{M}$ in this experiment. For each task-mode-worker cell we use 35 trajectories. The router is a deterministic prompt-only orchestrator evaluated under four nested pre-deployment records $Z_0, \dots, Z_3$, from budget-only context to full unmodified-artifact baseline. The primary decision metric is the first-passage deployment loss in (5); full-horizon occupancy and final quality are audit metrics. Appendix D gives the frozen configuration, including horizon, checker budget, training budget, success threshold, signal contents, measurement-cost convention, progress normalization, smoothing rule, and logging schema.

Experiment 1: Worker frontier. We run every worker on every task mode before routing, estimating fixed-setting analogues of $p(a, s)$ and $\kappa(a, s)$. These estimates instantiate the worker-side cost and competence terms in Theorem 1; the smoothed log-effort estimator is defined in Appendix D.

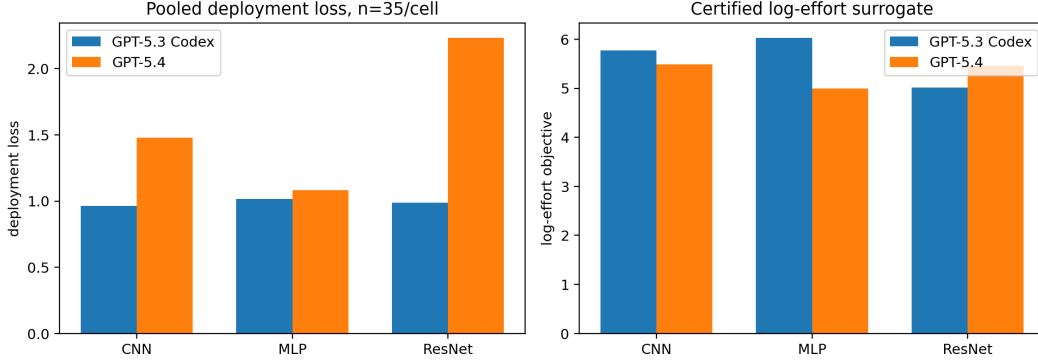


Figure 1: Two-worker frontier on the 35-run pooled panel. Left: first-passage deployment loss, lower is better. Right: certified log-effort surrogate. The two summaries disagree by mode: gpt_5_4 reaches the threshold faster and has higher final checked improvement, while deployment-loss accounting favors gpt_5_3_codex in the promoted two-worker panel.

Table 1: Different summaries select different workers on the same trajectories. Speed uses lower mean first-hit step; quality uses higher final relative improvement; lower is better for log-effort and deployment loss.

Criterion	<i>MLP-flat</i>	<i>compact CNN</i>	<i>micro ResNet</i>
First-hit speed	gpt_5_4	gpt_5_4	gpt_5_4
Final improvement	gpt_5_4	gpt_5_4	gpt_5_4
Certified log-effort	gpt_5_4	gpt_5_4	gpt_5_3_codex
Deployment loss	gpt_5_3_codex	gpt_5_3_codex	gpt_5_3_codex

269 The frontier already rejects a single leaderboard ordering. Averaged across modes, gpt_5_4 has lower
270 mean first-hit step (2.45 vs. 5.35), higher final improvement (0.309 vs. 0.206), and lower mean log-
271 effort (5.315 vs. 5.606). Under the declared deployment loss, however, always using gpt_5_3_codex
272 has lower mean loss than always using gpt_5_4 (0.989 vs. 1.598), and deployment-loss accounting
273 selects gpt_5_3_codex in all three modes. Table 1 makes the mismatch explicit: conventional
274 speed and quality favor gpt_5_4, the certified log-effort diagnostic is mode-dependent, and the
275 promoted deployment loss favors gpt_5_3_codex. This is the empirical role of the decomposition:
276 it exposes whether apparent performance comes from first-passage competence, cost, or mode-specific
277 comparative advantage before a deployment loss decides the recommendation. Appendix E provides
278 the threshold sensitivity, trajectory-level progress, cost, token, and router-record diagnostics behind
279 these aggregate summaries.

280 **Experiment 2: Signal audit.** We audit whether richer pre-deployment records change the de-
281 terministic router’s decision state. This is not an estimator of $I(S; Z_j)$: a Shannon-information
282 estimate would require a frozen finite representation or predictive model for the signal records. In
283 the current diagnostic, budget-only records recover the three task-mode labels at 0.349 accuracy,
284 while probe-containing records recover them perfectly on the diagnostic set. Thus the signals can
285 expose mode structure; the remaining question is whether the router converts that structure into better
286 actions.

287 **Experiment 3: Routing shift and action-frontier regret.** We measure this conversion with shift
288 rate and action-frontier regret. Shift rate is the fraction of modes for which the richer signal changes
289 the selected worker; action-frontier regret scores the selected action against the mode-conditional
290 frontier from Experiment 1. Both diagnostics are defined in Appendix D.

291 The final audit contains 480 prompt-only decisions. On real signal records, the router remains
292 strongly biased toward gpt_5_4. Mean log-effort regret is 0.177 for Z_0 , 0.183 for Z_1 , 0.148
293 for Z_2 , and 0.192 for Z_3 . Under the primary deployment loss, the non-deployable mode oracle
294 in Figure 2 coincides with always using gpt_5_3_codex. This makes the failure sharper than
295 missing a subtle switching rule: the prompt-only router remains biased toward gpt_5_4, the worker

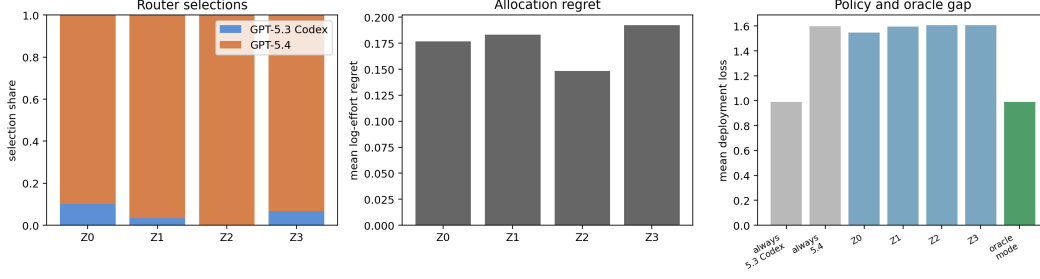


Figure 2: Two-worker router-response audit. Left: selected-worker distribution under real signal records. Middle: action-frontier regret against the 35-run pooled log-effort frontier. Right: deployment loss for always-worker baselines, prompt-router policies, and the mode oracle; under the primary loss the oracle coincides with always gpt_5_3_codex. Richer records expose mode information but do not reliably move the prompt-only router toward the deployment-loss recommendation.

Table 2: Paired router validation on the final two-worker menu. Net gain is $\hat{\ell}_{\text{dep}}(\rho_0) - \hat{\ell}_{\text{dep}}(\rho_j) - \hat{\ell}_{\text{meas}}(j)$; positive is better.

Signal	Pairs	Shift	Gross gain	Meas. loss	Net gain
Z_1	30	0.133	-0.049	0.000	-0.049
Z_2	30	0.100	-0.051	0.007	-0.059
Z_3	30	0.100	-0.032	0.028	-0.060

296 favored by speed and final-quality metrics rather than by the declared deployment loss. Under
 297 log-effort, the frontier remains mode-dependent, with gpt_5_4 favored on *MLP-flat* and *compact*
 298 *CNN* and gpt_5_3_codex favored on *micro ResNet*; the paired deployment loss decides which
 299 diagnostic should govern the promoted recommendation. Additional router-record completion and
 300 signal-ablation plots are reported in Appendix E.

301 **Experiment 4: Paired deployment gain.** For each richer signal Z_j , we compare the action
 302 selected under Z_j with the action selected under Z_0 on the same executable-setting panel and subtract
 303 measurement cost:

$$\hat{\Delta}_R(j) = \hat{\ell}_{\text{dep}}(\rho_0) - \hat{\ell}_{\text{dep}}(\rho_j) - \hat{\ell}_{\text{meas}}(j).$$

304 Positive values mean that the richer signal improves net deployment loss after the cost of measuring
 305 or using the signal is charged. This is the primary empirical estimand; positive signal information or
 306 lower regret alone is insufficient.

307 The paired validation is negative for the current prompt-only router. Against Z_0 , richer real signals
 308 produce small shift rates and negative net gain after measurement cost. The bootstrap intervals from
 309 the accounting script are $[-0.116, 0.002]$ for Z_1 , $[-0.127, -0.007]$ for Z_2 , and $[-0.113, -0.024]$
 310 for Z_3 . This is the main falsification result: informative records are not enough; the deployed router
 311 must use them in a way that improves the selected action’s checked cost–success tradeoff.

312 **Experiment 5: Negative controls.** We repeat the router and paired-gain analyses with schema-
 313 preserving controls: shuffled probe records, wrong-mode probes, and synthetic noise records
 314 with matched field names and approximate context burden. Controls are charged under the same
 315 measurement/context-cost convention as the corresponding real signal. They test whether router
 316 response and paired gain are specific to task-relevant signal content.

317 The controls block a strong routing-information claim. Wrong-mode Z_2 has a small positive mean net
 318 gain with an interval crossing zero, while several shuffled or synthetic controls are negative with inter-
 319 vals similar to the real signals. The correct conclusion is diagnostic rather than prescriptive: probes
 320 reveal task modes and the diagnostics expose ranking disagreement, but the primary deployment
 321 loss does not reward this router’s action changes in the promoted two-worker menu. Appendices D
 322 and E state the full promotion rule and provide the negative-control figure used to separate diagnostic
 323 improvements from a deployable routing recommendation.

6 Related work

The closest classical ancestor is algorithm selection. Since Rice’s formulation, solver portfolios have treated performance as instance-dependent: SATzilla and AutoFolio learn to select solvers from instance features rather than report one globally best solver [5–8]. Our contribution is the specialization to externally checked LLM agents, where a router observes a controlled pre-deployment record and is scored by paired deployment loss rather than standalone routing accuracy. Unlike AutoML systems such as auto-sklearn, which tune pipelines offline for a dataset, our unit is a deployable action for a concrete verifier-backed instance [9].

LLM routing and cascades provide the model-menu context. FrugalGPT, RouteLLM, RouterBench, LLMRouterBench, and compound-system routers study cost-quality routing over heterogeneous model pools [10–16]. We ask how much checked, task-specific information is worth before committing to an expensive agentic deployment.

Checked agent benchmarks provide the task substrate. SWE-bench evaluates repository repair against tests, AutoResearch evaluates training-loop edits against validation feedback, and theorem-proving systems use analogous external validators [1, 17]. Repeated sampling and inference-time scaling further show that lower-cost or smaller models can become competitive when verification is available [18–21]. Our question is upstream of the launched worker: which action should be selected, what should the router observe, and how should that observation be charged?

7 Discussion and limitations

The paper separates a theory-facing diagnostic claim from a deployment-facing validation claim. The experiments support the diagnostic claim: first-passage speed, final improvement, certified log-effort, and deployment loss disagree on the same checked task family. They do not support the stronger claim that the evaluated prompt-only router improves deployment loss. Probe records expose the predeclared task modes, but under the primary deployment loss the mature two-worker menu does not create a positive switching opportunity beyond always using `gpt_5_3_codex`; the router remains biased toward the worker favored by speed and final-quality metrics.

The main limitations are the packed-mode abstraction, the fixed executable prototypes, and the two-worker mature action menu. Real agent trajectories are stateful, task modes may be fuzzy, and the reported quantities are fixed-panel diagnostics rather than population-level deployment estimates. This is why the appendix reports threshold curves, occupancy, cost sensitivity, token accounting, and signal-ablation checks instead of treating the finite-mode surrogate as sufficient. The decomposition is therefore a claim discipline: it helps identify cost, competence, information, and allocation mechanisms, but paired deployment gain decides whether a routing recommendation is deployable.

Future work. The immediate experimental extension is to add a stable lower-cost action with matched trajectory support. That would make the worker menu span a broader cost–capability frontier than the two-worker comparison reported here. It would also test the retry-crossover theorem in its intended regime, where a lower-cost worker may compensate for lower one-attempt success through repeated checked attempts. The ongoing `gpt_5_4_mini` cells are not promoted here because support is still uneven across modes; they are the natural candidate for the retry-crossover experiment once matched at 35 trajectories per cell. Beyond this benchmark, repository repair, theorem checking, and multi-file research-code editing would test less artificial mode schemas and more heterogeneous measurement costs.

Conclusion. Model choice inside an agent harness is not a scalar leaderboard problem. It is a budget-aware deployment decision validated by paired outcomes after measurement cost. The final two-worker study asks that question directly and answers it negatively for the current prompt-only router: pre-deployment information does not move allocation reliably enough to improve paired deployment loss.

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A Proofs and retry rule

A.1 Four-term identity

Under Assumption 1,

$$\log T_{\rho,q}(S, Z) = C + \log \kappa(a_Z, S) - \log p(a_Z, S) - \log q_Z(S).$$

Conditioning on $Z = z$,

$$\mathbb{E}[-\log q_z(S) \mid Z = z] = \sum_s \pi_z(s) \log \frac{1}{q_z(s)} = H(\pi_z) + \text{KL}(\pi_z \parallel q_z).$$

Therefore

$$\mathbb{E}[\log T_{\rho,q}] = C + \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} [\log \kappa(a_Z, S) - \log p(a_Z, S)] + \mathbb{E}_Z [H(\pi_Z)] + \mathbb{E}_Z [\text{KL}(\pi_Z \parallel q_Z)].$$

For the prior-matched baseline ρ_0 that deploys fixed action a_0 and uses $q_0(\cdot \mid z) \equiv \pi$,

$$\mathbb{E}[\log T_{\rho_0, \pi}] = C + \mathbb{E}_{S \sim \pi} [\log \kappa(a_0, S) - \log p(a_0, S)] + H(\pi).$$

Subtracting the deployed effort from the baseline effort yields

$$\Delta = \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[\log \frac{\kappa(a_0, S)}{\kappa(a_Z, S)} \right] + \mathbb{E}_Z \mathbb{E}_{S \sim \pi_Z} \left[\log \frac{p(a_Z, S)}{p(a_0, S)} \right] + H(\pi) - \mathbb{E}_Z [H(\pi_Z)] - \mathbb{E}_Z [\text{KL}(\pi_Z \parallel q_Z)].$$

The entropy difference is $I_\pi(S; Z)$, giving (4).

A.2 Pilot concentration

For binary success observations in an action-mode cell, Hoeffding gives

$$\mathbb{P}(|\hat{p}(a, s) - p(a, s)| > \alpha) \leq 2 \exp(-2n_0\alpha^2).$$

If all true probabilities are at least $p_{\min}$, then the log-probability error obeys $|\log \hat{p} - \log p| \leq \alpha/p_{\min} + O(\alpha^2/p_{\min}^2)$ on the good event. A union bound over $|\mathcal{A}| |\mathcal{S}|$ cells yields the conservative pilot rule

$$n_0 = O\left(\frac{\log(|\mathcal{A}| |\mathcal{S}| / \delta_{\text{conf}})}{\delta_{\text{acc}}^2 p_{\min}^2}\right).$$

Wilson intervals and bootstrap intervals are reported in practice because pilot cell counts are small.

B Operational diagnostics

The identity above becomes useful only after each term is tied to quantities logged by the deployment protocol. The checker-facing progress process is the common object used to operationalize competence, cost, information, and allocation diagnostics. Fix a horizon H and a task-normalized checked progress process $G_t(a, x) \in [0, 1]$ for action a on instance x . In the main experiment, the router acts

once before deployment: after observing Z , it selects one action, and that action is used for the whole trajectory. The success and cost diagnostics below are therefore trajectory-level quantities computed from the step-wise verifier trace. For lower-is-better checker losses, one admissible normalization is

$$G_t(a, x) = \text{clip}_{[0,1]} \left(\frac{L_0(x) - L_t(a, x)}{|L_0(x)| + \varepsilon_L} \right),$$

where $L_0(x)$ is the starting-artifact loss and $L_t(a, x)$ is the checked loss convention at step t . Given a success threshold $\delta > 0$,

$$\tau_\delta(a, x) = \inf\{t \leq H : G_t(a, x) \geq \delta\}, \quad F_\delta(a, x) = \mathbf{1}\{\tau_\delta(a, x) = \infty\}.$$

We also log persistence through

$$\text{Occ}_{\delta,H}(a, x) = \frac{1}{H} \sum_{t=1}^H \mathbf{1}\{G_t(a, x) \geq \delta\}.$$

Competence. The action competence component $p(a, s)$ is the first-success probability of action a on task mode s :

$$p(a, s) = \mathbb{P}[\tau_\delta(a, X) \leq H \mid S = s].$$

Thus G_t is step-wise, but the policy competence term uses the probability that the worker selected by the router reaches the success threshold at least once before the horizon. Cumulative progress and persistence are not substituted for $p(a, s)$ in the identity. We log

$$\bar{G}_a(s) = \mathbb{E} \left[\frac{1}{H} \sum_{t=1}^H G_t(a, X) \mid S = s \right]$$

and $\text{Occ}_{\delta,H}$ as secondary diagnostics, especially when first-hit success saturates or when partial progress is scientifically relevant.

Cost. The action cost component $\kappa(a, s)$ is kept separate from competence. In the main protocol it is the normalized cost of deploying action a on mode s for the same trajectory convention used to define $p(a, s)$. The routed-policy cost term averages this mode-conditional cost over the actions selected under Z . The primary convention is cost accumulated until the first certified success or the horizon, $\tau_\delta \wedge H$. We log

$$C_\delta(a, x; \xi) = \lambda^\top (\tilde{T}_{\tau_\delta \wedge H}^{\text{worker}}, \tilde{C}_{\tau_\delta \wedge H}^{\text{tok}})(a, x; \xi),$$

and report worker wall-clock cost as primary, with token count as a sensitivity analysis. Checker runtime is logged for audit, but it is not part of $\kappa(a, s)$ unless the checker protocol or checker-call frequency differs across actions.

Routing information. The theorem term $I(S; Z)$ measures how much a router-visible signal Z separates the predeclared task modes before deployment. This does not mean that the evaluator already knows the best worker for each instance. The evaluator knows only the candidate mode schema and can estimate whether Z_j makes that schema more identifiable:

$$I(S; Z_j \mid Z_0) = H(S \mid Z_0) - H(S \mid Z_j).$$

The quantity is estimated only when a finite signal representation or calibrated predictive model for the signal records is predeclared. Otherwise the paper reports signal summaries and router-response diagnostics instead of a Shannon mutual-information estimate. It is a descriptive information diagnostic; its decision value is established only after action outcomes show that the modes predict different cost–success profiles. The paired router experiment then checks whether higher-information signals actually change the selected action and improve first-passage deployment outcomes.

Allocation-summary mismatch and allocation regret. The theorem term $\mathbb{E}_Z \text{KL}(\pi_Z \| q_Z)$ is defined only after the evaluator predeclares a reproducible construction of the evaluator-side mode summary q_Z . Operationally, the router returns an action, not a mode label or distribution. When no such q_Z construction is frozen, the experiment reports allocation regret as a separate action-frontier diagnostic. After deployment runs, the evaluator estimates the mode-conditional frontier

$$a^*(s) \in \arg \min_{a \in \mathcal{A}} \{\log \kappa(a, s) - \log p(a, s)\},$$

or the statistically indistinguishable set of frontier actions. For a selected action $a = \rho_j(x)$ and mode $s = \sigma(x)$, the corresponding allocation regret diagnostic is

$$r_j(x) = [\log \kappa(a, s) - \log p(a, s)] - \min_{a' \in \mathcal{A}} [\log \kappa(a', s) - \log p(a', s)].$$

This regret is computed retrospectively on a diagnostic split and validated on holdout instances; it is not an oracle input to the router. High information with high allocation regret means that the progressive signal exposed useful mode structure, but the induced action did not exploit the empirical cost–success frontier.

After these diagnostics are estimated, the experiment still reports the decision-facing first-passage deployment loss in (5) as final validation. The full-horizon audit loss in (6) is reported separately for persistence and final-quality checks. These losses are not replacements for the theorem; they test whether designs favored or explained by the diagnostics improve the practical deployment objective.

C Progressive-signal decomposition

The main four-term identity compares a deployed design to a prior-matched baseline. For the V3 router experiment, it is often useful to compare two progressive signal conditions for the same router. Assume a finite task mode $S \in \mathcal{S}$ and, for each signal condition j , the conditional mode distribution induced by the realized signal record

$$\pi_j(s | Z_j) = \mathbb{P}[S = s | Z_j].$$

The fixed orchestrator R induces the routing policy $a_R^j(Z_j) = R(Z_j)$. If an evaluator-side mode summary $q_R^j(\cdot | Z_j) \in \Delta(\mathcal{S})$ has been predeclared, define

$$\mathcal{E}_R^j(s, Z_j) = \frac{\kappa(a_R^j(Z_j), s)}{p(a_R^j(Z_j), s) q_R^j(s | Z_j)}.$$

The log-effort gain of signal condition Z_j over Z_0 is

$$\Delta_R^{\log}(j) := \mathbb{E}[\log \mathcal{E}_R^0(S, Z_0) - \log \mathcal{E}_R^j(S, Z_j)].$$

Expanding the cross-entropy terms gives

$$\Delta_R^{\log}(j) = C_R(j) + \Phi_R(j) + G(j) + M_R(j),$$

where

$$\begin{aligned} C_R(j) &= \mathbb{E} \left[\log \frac{\kappa(a_R^0(Z_0), S)}{\kappa(a_R^j(Z_j), S)} \right], \\ \Phi_R(j) &= \mathbb{E} \left[\log \frac{p(a_R^j(Z_j), S)}{p(a_R^0(Z_0), S)} \right], \\ G(j) &= H(S | Z_0) - H(S | Z_j), \\ M_R(j) &= \mathbb{E} \left[\text{KL}(\pi_0(\cdot | Z_0) \| q_R^0(\cdot | Z_0)) - \text{KL}(\pi_j(\cdot | Z_j) \| q_R^j(\cdot | Z_j)) \right]. \end{aligned}$$

When Z_j refines Z_0 , $G(j) = I(S; Z_j | Z_0)$. This signal-mode decomposition is theory-facing and excludes measurement cost unless such a term is added explicitly. It is a diagnostic companion to the deployment-gain estimand (8), not a replacement for it.

D Full selection protocol and experimental configuration

Panel of fixed executable settings. The main experiment is a repeated-trajectory panel of fixed executable settings rather than a population draw from μ . Each task mode s is represented by one executable setting

$$\chi_s = (A_s, D, \text{split}, u, \eta_s, V, B_V, H, \delta),$$

where A_s is the editable `train.py` template and starting model class, D is CIFAR-10, `split` is the fixed train/validation split, u is the model-initialization seed, η_s are starting optimization and architecture hyperparameters, V is the checker, B_V is the checker budget, H is the edit–verify horizon, and δ is the success threshold. Panel-level aggregates use $\hat{\pi}(s) = 1/3$. The analyzed worker menu is `{gpt_5_3_codex, gpt_5_4}`. The paper-facing comparison sets $\mathcal{A} = \mathcal{M}$: each deployable action is a single worker run under the common prompt and harness.

Task-mode construction. The paper-facing task modes are named by the starting model family in the editable artifact. *MLP-flat* (`mlp_flat`) is a fully connected classifier over flattened CIFAR-10 inputs; *compact CNN* (`cnn_compact`) is a small convolutional classifier with local spatial inductive bias; and *micro ResNet* (`resnet_micro`) is a reduced residual convolutional classifier with skip connections. They are not different datasets or different workers. They are evaluator-defined variants of the same CIFAR-10 edit-verify task: the dataset, split, checker, prompt harness, horizon, and success threshold are fixed, while the initial architecture family and associated template parameters change. This choice makes the mode variable operationally meaningful for the theory, because different starting artifacts can change both first-passage competence $p(a, s)$ and cost $\kappa(a, s)$ for the same worker.

Progressive signal records. The router sees one of four nested pre-deployment records. Z_0 contains only the task name, checker objective, worker menu, and budget. Z_1 adds the initial `train.py` plus a static workload summary: starting model class, parameter count, optimizer and budget settings, dataset/checker identifiers, and parse/runtime availability. Z_2 adds a low-cost checker probe of the unmodified artifact at a predeclared probe budget. Z_3 adds the full unmodified-artifact baseline under the deployment checker budget. Probe records include validation loss, validation accuracy, training seconds, total seconds, optimizer steps, parameter count, parse/runtime status, and probe budget. Probe runs are outside the selected worker deployment: they are charged through ℓ_{meas} and do not consume the later deployment horizon or checker budget. The evaluator-only baseline used to compute G_t is router-visible only in Z_3 .

Experiment accounting. The worker frontier uses 35 trajectories per task-mode-worker cell. The selected-router validation uses the same fixed panel and reports all richer signals $j \in \{1, 2, 3\}$ rather than selecting a single best signal on the same data. Negative controls keep the signal schema and approximate context burden of the matched real signal: shuffled probes, wrong-mode probes, and synthetic noise records are charged under the same measurement/context-cost convention as the real record they mimic. For the router audit, the prompt-only router selects `gpt_5_4` on 27/30 budget-only records, 29/30 static-workload records, 30/30 probe records, and 28/30 full-scout records.

Promotion rule. A routing recommendation is promoted only when worker-frontier estimates, cost sensitivity, router shift, action-frontier regret, paired gain after measurement cost, and negative controls agree. If diagnostics improve but paired deployment gain does not, the report treats the result as a mechanism failure rather than a useful router.

Trajectory record. Each run stores the worker alias, resolved model identifier, API date, split, task family, task mode, seed, maximum horizon H , signal condition, prompt hash, router model identifier, router temperature/top-p, router seed policy, parser version, router output, selected worker, and cost convention. Each step stores the prompt-visible state, proposed `train.py` artifact, parse status, checker result, validation loss, validation accuracy, relative improvement, incremental worker wall time, token usage, and checker runtime for audit. Parser failures are logged as unsuccessful routing or deployment records under the predeclared parser-failure policy.

Success and time-to-verification. For trajectory i , τ_i is the first step at which the relative improvement threshold is crossed; censored failures have $\tau_i = \infty$. For each action-mode cell,

$$\hat{F}_a(s, h) = \frac{1}{n_{a,s}} \sum_i \mathbf{1}\{\tau_i \leq h\}, \quad \hat{S}_a(s, h) = 1 - \hat{F}_a(s, h).$$

The empirical hazard is

$$\hat{\lambda}_a(s, h) = \frac{\sum_i \mathbf{1}\{\tau_i = h\}}{\sum_i \mathbf{1}\{\tau_i \geq h\}}.$$

The raw first-success estimate is $\hat{p}(a, s) = \hat{F}_a(s, H)$. For log-effort scores, the protocol uses the predeclared smoothed estimate

$$\tilde{p}(a, s) = \frac{k_{a,s} + 1/2}{n_{a,s} + 1},$$

where $k_{a,s}$ is the number of first successes in the cell. This prevents zero-success pilot cells from making log scores infinite; raw success counts are still reported.

Table 3: Frozen operational configuration for the main protocol. All values are fixed within a task mode; repeated runs vary only the agentic trajectory.

Component	Fixed value
Dataset and split	CIFAR-10, 50,000 training examples and 10,000 validation examples; balanced subset, label noise 0.0, imbalance ratio 1.0; the train/validation split is fixed in the main protocol
Data seed	Common fixed data/subset seed; alternative split seeds are reported only as robustness variants
<code>train.py</code> initialization seed	SEED=42; used for Python, NumPy, PyTorch, CUDA seeds, deterministic PyTorch mode, and the training-loader generator; alternative initialization seeds are robustness variants
Training transforms	random crop with padding 4, random horizontal flip, CIFAR-10 normalization; validation uses normalization only
Training budget per checker call	fixed-step mode with <code>A_MAX_STEPS</code> =256; time budget 120s is a fallback, not the primary scoring convention
Common hyperparameters	<code>BATCH_SIZE</code> =64, <code>NUM_WORKERS</code> =0, Adam, learning rate $5 \cdot 10^{-4}$, weight decay 10^{-4} , betas (0.9, 0.999), no LR schedule
Task mode	<i>MLP-flat</i> (<code>mip_flat</code>), <i>compact CNN</i> (<code>cnn_compact</code>), or <i>micro ResNet</i> (<code>resnet_micro</code>)
Shared template parameters	depth 2, base channels 12, channel multiplier 2, batch norm enabled, dropout 0.0, and hidden width 48
Agent horizon	$H = 20$ edit-verify steps; trajectories continue to H after first success to measure persistence and final quality
Progress normalization	$\varepsilon_L = 10^{-8}$ in the denominator of G_t
Primary loss weights	$\alpha_{\text{fail}} = 1$; normalized worker wall-clock is the primary scalar C_δ ; token cost is a sensitivity analysis
Audit loss weights	$\alpha_{\text{occ}} = \alpha_{\text{qual}} = 1$, $\psi(u) = 1 - u$
Measurement cost	pre-deployment probes are charged through ℓ_{meas} using the same wall-clock normalizer as C_δ ; context-token cost is reported as sensitivity
Router configuration	deterministic prompt-only router with fixed model identifier, prompt hash, parser, temperature, top-p, and seed policy recorded before evaluation

Cost. The primary cost is cumulative worker wall-clock time to $\tau \wedge H$. Secondary worker-side cost is cumulative token count. Checker runtime is logged as a protocol audit field and excluded from action-cost comparisons unless the checker protocol differs across actions. All plots that support model recommendations must be reproducible under at least worker wall-clock and token-count conventions. When a worker fails before producing a parseable artifact, its cost is still counted and the step is marked unsuccessful. The scalar estimator matched to Assumption 1 is the cell mean

$$\hat{\kappa}(a, s) = \frac{1}{n_{a,s}} \sum_{i: S_i=s} C_{\delta,i}(a),$$

or its fixed-setting analogue $\hat{\kappa}_{\chi_s}(a)$ in Section 5.

Information, allocation summaries, and regret. For each progressive signal Z_j , a Shannon information estimate is reported only if the protocol predeclares a finite signal representation or calibrated predictive model for estimating $\pi_j(\cdot | Z_j)$ on held-out instances. Otherwise the report uses qualitative signal summaries and router-response diagnostics without calling them estimates of $I(S; Z_j)$. The KL mode-summary divergence term is reported only if the evaluator also freezes a reproducible construction of $\hat{q}_R^j(\cdot | Z_{j,i})$ before seeing evaluation outcomes. For the deterministic router, the empirical panel shift rate is

$$\widehat{\text{Shift}}_R(j) = \sum_{s \in \mathcal{S}} \hat{\pi}(s) \mathbf{1}\{\rho_j(\chi_s) \neq \rho_0(\chi_s)\}.$$

In the main fixed-setting protocol, the primary operational diagnostic is allocation regret,

$$\hat{r}_j(x_i) = [\log \hat{\kappa}(a_{j,i}, s_i) - \log \tilde{p}(a_{j,i}, s_i)] - \min_{a \in \mathcal{A}} [\log \hat{\kappa}(a, s_i) - \log \tilde{p}(a, s_i)],$$

where $a_{j,i} = R(Z_j(x_i))$ and $s_i = \sigma(x_i)$.

Uncertainty. Success probabilities use Wilson intervals. Aggregated objectives, paired gains, and calibration diagnostics use bootstrap or randomization tests at the sampling unit declared for the study. If the main protocol remains a panel of fixed executable settings, paired randomization tests over repeated trajectories are descriptive for that panel and should not be interpreted as population-level inference. If independent task instances are added, uncertainty is reported at the task-instance level.

E Additional plots

This appendix collects auxiliary diagnostics generated from the experimental logs. The threshold, trajectory, and cost diagnostics below use the same analyzed workers, `gpt_5_3_codex` and `gpt_5_4`. These appendix plots use the same 35-trajectory-per-cell panel as the main analysis. The pooled appendix plots are robustness and mechanism diagnostics for the same accounting quantities reported in the main text.

Table 4: Worker-frontier summary on the 35-run pooled panel. Each cell uses 35 trajectories. Lower is better for log-effort and deployment loss.

Mode	Worker	Success	Mean τ	Log-effort	Dep. loss
<i>MLP-flat</i>	<code>gpt_5_3_codex</code>	32/35	7.41	6.029	1.015
<i>MLP-flat</i>	<code>gpt_5_4</code>	35/35	2.49	4.991	1.083
<i>compact CNN</i>	<code>gpt_5_3_codex</code>	34/35	6.03	5.774	0.964
<i>compact CNN</i>	<code>gpt_5_4</code>	35/35	3.06	5.492	1.479
<i>micro ResNet</i>	<code>gpt_5_3_codex</code>	35/35	2.60	5.015	0.987
<i>micro ResNet</i>	<code>gpt_5_4</code>	35/35	1.80	5.461	2.234

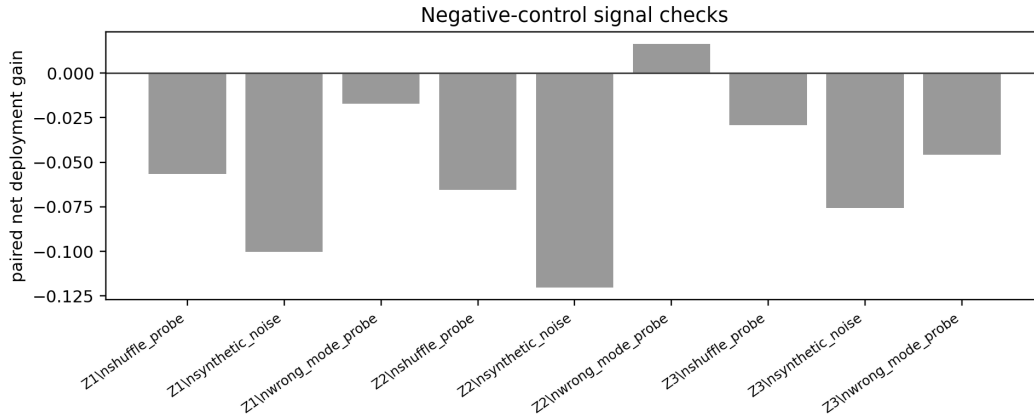


Figure 3: Negative-control paired gains for shuffled probes, wrong-mode probes, and synthetic noise records. The real records do not cleanly outperform the controls, indicating that the current prompt-only router is not reliably exploiting task-relevant measurement content.

E.1 Pooled threshold and trajectory diagnostics

E.2 Pooled cost diagnostics

Threshold sensitivity, two mature workers pooled common support (n=210)

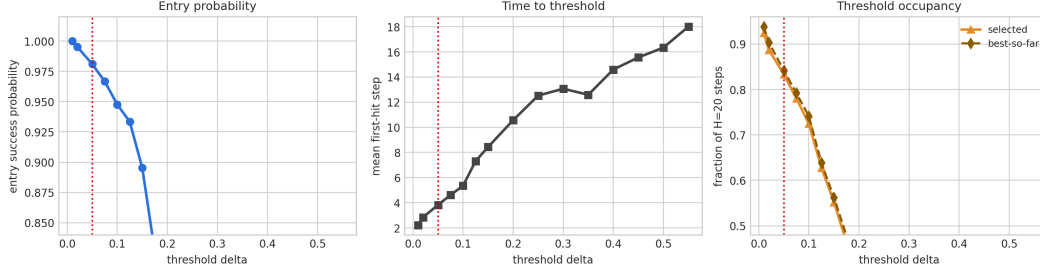


Figure 4: Zoomed threshold sensitivity for the two mature workers on the 35-run pooled panel. The red line marks the operating threshold $\delta = 0.05$. At this threshold, entry success is near saturated while mean first-hit time remains low, which makes $\delta = 0.05$ a pragmatic operating point for studying time-to-certified-solution. The sharp drop at larger thresholds shows why the exact threshold is not an innocuous plotting choice.

Extended threshold sensitivity by worker, pooled common support (n=210)

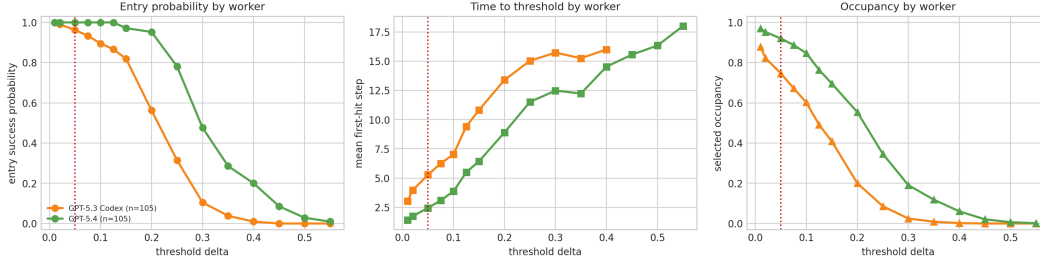


Figure 5: Threshold sensitivity by worker on the 35-run pooled panel. gpt_5_4 remains stronger at higher thresholds, while gpt_5_3_codex degrades earlier; this supports the competence term in Theorem 1.

Extended threshold sensitivity, two mature workers pooled common support (n=210)

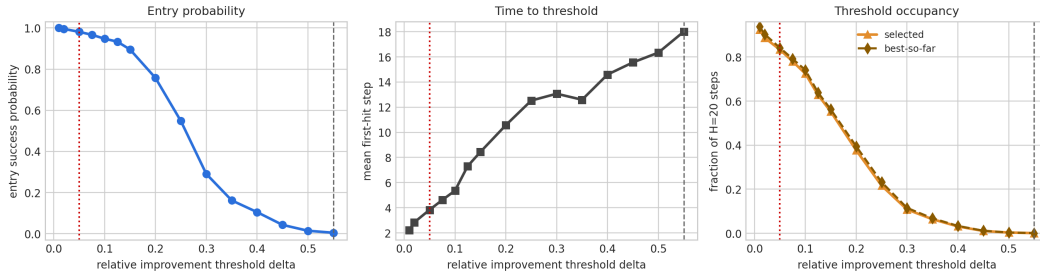


Figure 6: Extended threshold sensitivity for the two-worker pooled panel. The full threshold range shows that high-threshold success is rare even when $\delta = 0.05$ is nearly saturated. This is why the paper reports first-passage curves, occupancy, and final quality rather than relying on a single binary success rate.

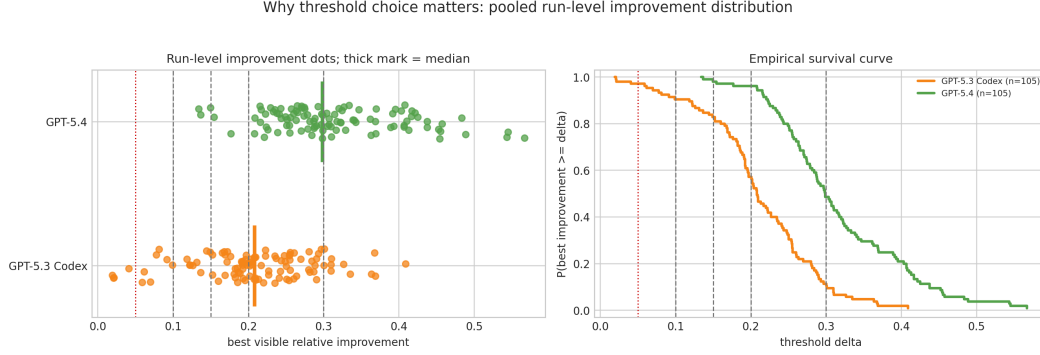


Figure 7: Run-level improvement distribution for the two mature workers. Dots show final best-visible relative improvement and thick ticks mark worker medians. The survival plot makes the threshold choice explicit: near $\delta = 0.05$, most runs succeed, but the ordering and separation change at higher thresholds. This diagnostic supports the paper’s claim that final quality, first-passage success, and deployment loss are distinct empirical objects.

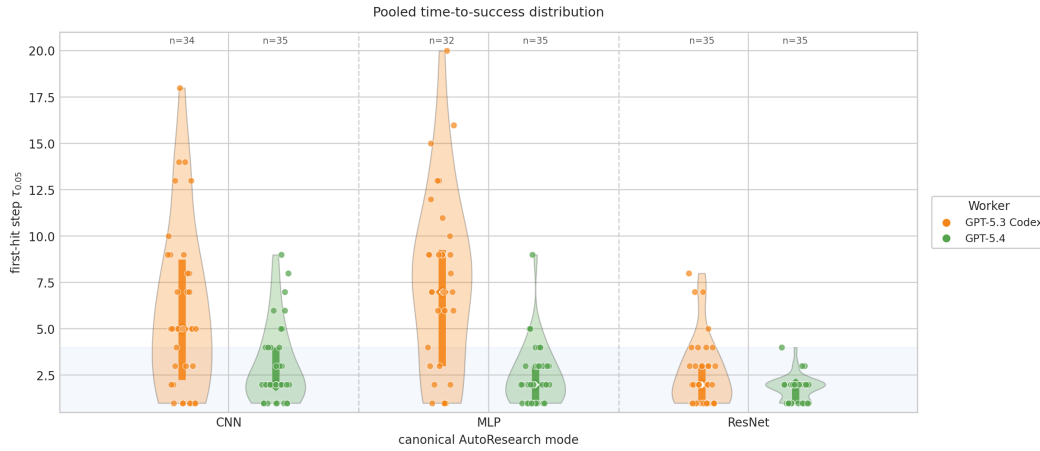


Figure 8: Pooled time-to-success distribution by task mode and worker. The violin plots expose mode-conditional heterogeneity: gpt_5_4 typically hits the threshold quickly, while gpt_5_3_codex has broader first-hit times, especially on MLP. This is the empirical object behind the cost and competence terms of the four-term decomposition.

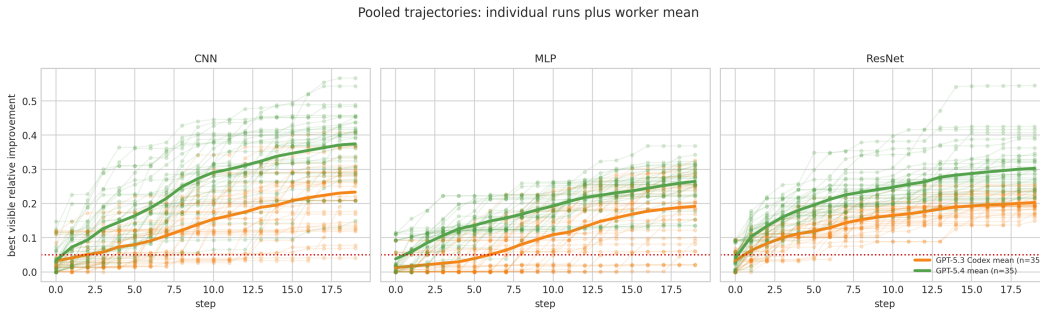


Figure 9: Best-so-far improvement trajectories. The mean trajectories show that gpt_5_4 improves earlier and reaches higher final relative improvement in the pooled panel, but the deployment accounting in the main text shows that this does not automatically imply lower deployment loss after cost is charged. This contrast is the main empirical reason to separate worker competence from deployment value.

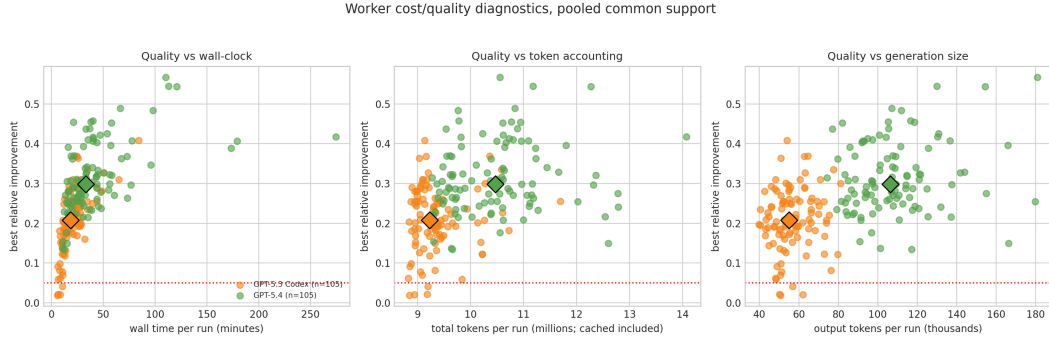


Figure 10: Worker cost–quality diagnostics. Each point is a pooled common-support run, and diamonds mark worker medians. The plots show that better final improvement is accompanied by substantial variation in wall-clock time, total token accounting, and generated-token volume. This supports the paper’s budget-aware framing: worker choice cannot be reduced to final improvement alone.

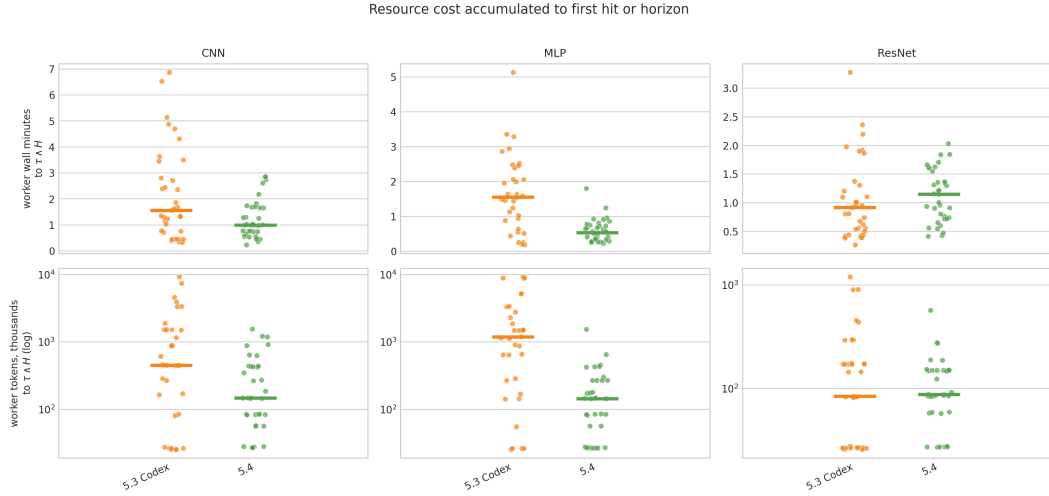


Figure 11: Resource cost accumulated to first hit or horizon. The top row reports wall-clock minutes and the bottom row reports token accounting on a log scale. Mode-conditional cost spreads are large, which is why the retry-crossover theorem is treated as a cost-sensitive design rule rather than a generic endorsement of weaker workers.

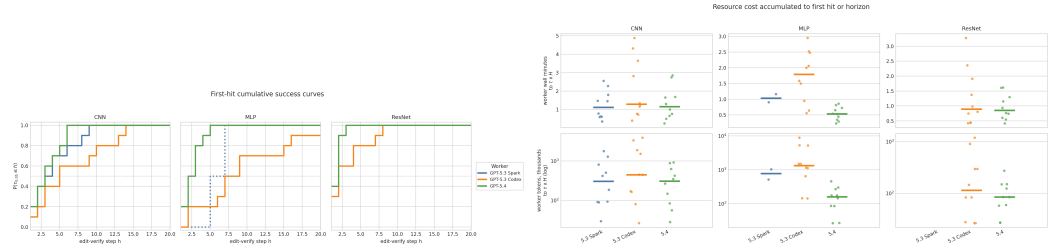


Figure 12: First-passage diagnostics. Left: first-hit CDFs by mode. Right: cost-to-threshold by mode and worker. These are the direct empirical objects behind the retry and time-to-certified-solution interpretation.

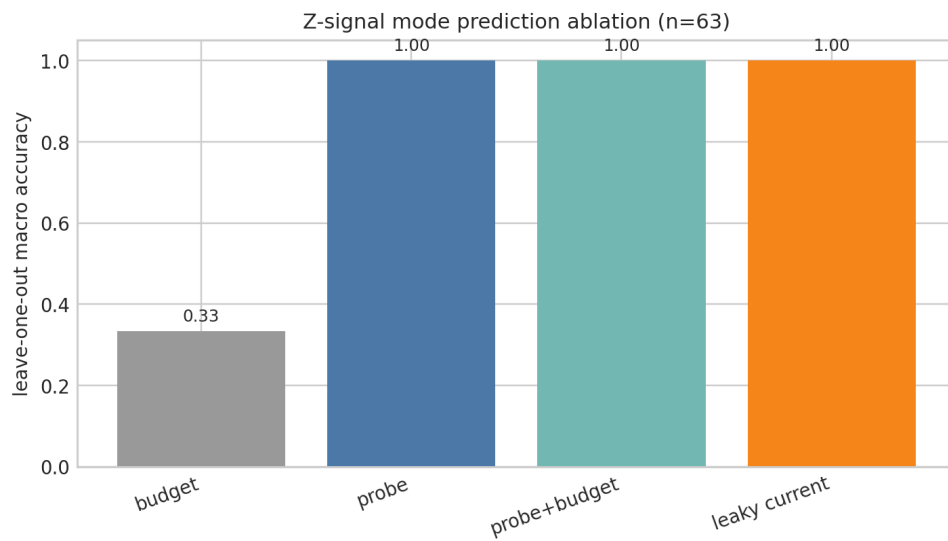


Figure 13: Router-facing signal ablation. The diagnostic tests whether progressively richer records expose the task-mode structure used in the accounting analysis.

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2. Limitations

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Justification: Section 5 describes the AutoResearch task instantiation, task modes, model menu, router-visible signals, horizons, metrics, and evaluation protocol; Appendix D specifies the required logs, estimators, and run-metadata fields.

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707 logging, and routing-output requirements.